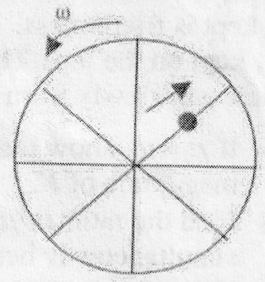
PH-251 [M]

Tutorial-I

Newtonian Mechanics, Frictional forces, Inertial and Non-inertial frames

Inertial frame

1 A bead moves with a speed u = (1+0.5t) along the spoke of a wheel as shown in the figure. The wheel rotates with an angular velocity ω = (1+0.5t). The bead is at the center of the wheel and the spoke is along x-axis at $t = 0$. Using the plane polar coordinate, find the velocity and the acceleration of the bead at $t = 2$ s.



[Ans: $\vec{v} = 2\hat{r} + 6\hat{\theta}$; $\vec{a} = -11.5\hat{r} + 9.5\hat{\theta}$]

*2. An insect starts from the center of a rotating turntable at time $t = 0$ with a constant speed of 0.03 m/s along a radial line. At any instant of time t, the angular displacement θ of this line is given by $\theta = \frac{\pi}{4} + 6t - 3t^2$; where θ is in radians and t is in seconds. With respect to an inertial observer and using the plane polar coordinate system answer the following:

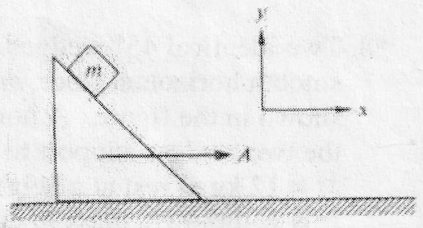
(a) Write down the acceleration of the insect as a function of time,

(b) At what time the acceleration of the insect is purely radial? What is the value of this acceleration.

(c) At what time the acceleration of the insect is purely tangential? What is the value of this acceleration?

[Ans: 2/3 s, 1 s]

3. A 45° wedge is pushed along a table with constant acceleration A. A block of mass m slides without friction on the wedge. Find its acceleration. (Gravity is directed down).

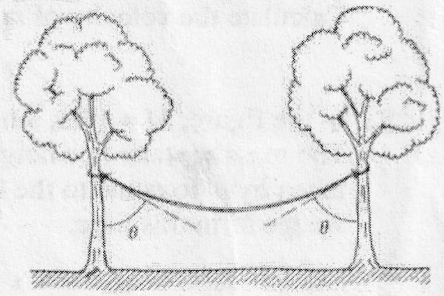


[Ans: $a_x = \frac{A+g}{2}$; $a_y = \frac{A-g}{2}$]

4. A uniform rope of weight W hangs between two trees. The ends of the rope are at the same height, and they each make angle θ with the trees. Find

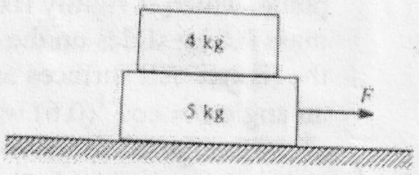
(a) The tension at either end of the rope.

(b) The tension in the middle of the rope.



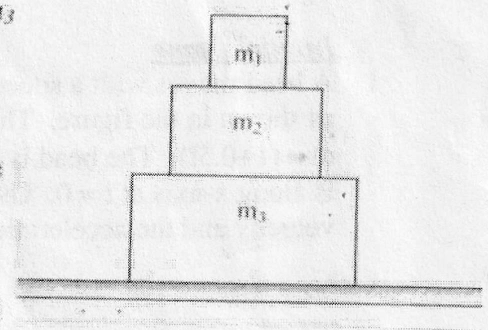
[Ans: $T_{middle} = \frac{W\tan\theta}{2}$; $T_{end} = \frac{W\sec\theta}{2}$]

5. A 4 kg block rests on top of a 5 kg block, which rests on a frictionless table. The coefficient of friction between the two blocks is such that the blocks start to slip when the horizontal force F applied to the lower block is 27 N. Suppose that a horizontal force is now applied only to the upper block. What is its maximum value for the blocks to slide without slipping relative to each other?



[Ans: 21.6 N]

*6. Three masses m_1, m_2 and m_3 are placed, one on the top of other as shown in the figure. The coefficient of friction between m_1 and m_2 is μ_1, while that between m_2 and m_3 is μ_2. The surface on which m_3 is kept is frictionless. A horizontal force F is applied to the mass m_1 kept on the top. The force F is applied in such a way that it increases slowly from zero.

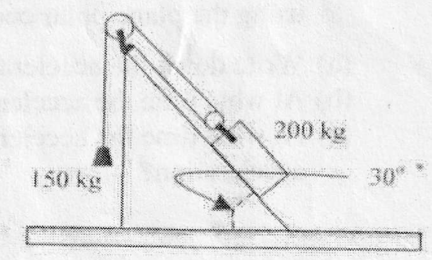


a) If $\mu_1=\mu_2$, show that m_2 would not slip on m_3 irrespective of the magnitude of F.

b) Find the ratio μ_1/μ_2, for which slipping would start simultaneously between m_1, m_2 and m_2, m_3.

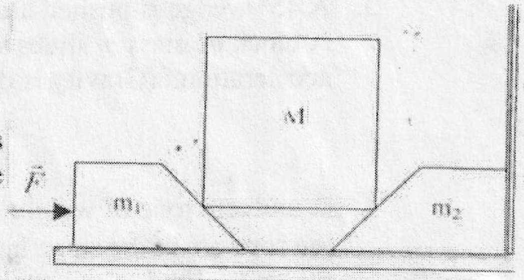
c) If the ratio μ_1/μ_2 is more than the one obtained in part (b), find the magnitudes of the forces for which slipping would start between m_1, m_2 and between m_2, m_3. Also plot the accelerations of three masses as a function of magnitude of F. Consider f_1 is the value of friction between m_1 and m_2 and f_2 the value of it between m_2 and m_3, plot f_1 and f_2 as a function of magnitude of F.

7. A crate of mass 200 kg is being raised up a 30° incline the way shown in the figure below. If the coefficient of friction μ between the crate and the incline is 0.5, calculate the acceleration of the crate.



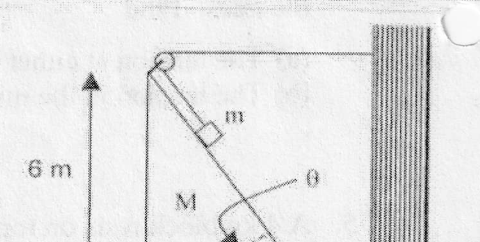
[Ans: 1.4 m/s^2]

*8. Two identical ~~45~~30° wedges m_1 and m_2 of mass 1 kg each, rest on a smooth horizontal floor, m_2 being held against a vertical wall as shown in the figure. A horizontal force F is applied to m_1 so that the two wedges support to keep a symmetric smooth surfaced mass $M = 12$ kg at rest at a height of 1 m from the floor. When the force F is withdrawn, mass M comes down pushing m_1 to the left. Calculate the velocity of m_1, 3 s after F is withdrawn.



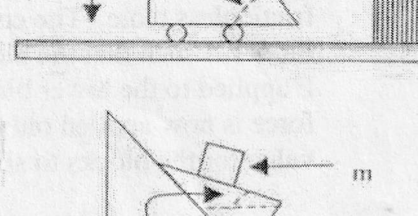
[Ans: 6.8 m/s]

9. In the figure, $M = 16m$, $\sin\theta = 0.8$. All surfaces are frictionless. The mass m starts at a height of 6 m from the floor. Find the time taken by m to come to the floor and the distance moved by the wedge M in this time.



[Ans: 6.06 s, 7.5 m]

10. A triangular wedge (M) of mass 1.2 kg slides along an inclined plane, which is rigidly fixed to the ground. A small block (m) of mass 0.6 kg slides on the upper surface of the wedge as shown in the figure. All surfaces are frictionless. The inclined plane makes an angle $\theta = \cos^{-1}(0.6)$ with the horizontal, while the upper surface of the wedge on which m slides makes an angle $\phi = \cos^{-1}(0.8)$ with the horizontal. Obtain the horizontal and vertical components of the acceleration of m and M.



[Ans: horizontal component of acceleration of m =144g/433]

Non-inertial frame

*11.A body of mass 1 kg is kept on the floor of a moving truck at a distance of 3 m from its rear end. The coefficient of static friction between the floor and the body is 0.24 while that of kinetic friction is 0.18. (a) Find the maximum value of the acceleration a_{max} that the truck may have in the forward direction for which the body does not slip on the floor.
(b) Suppose the truck is given an initial acceleration slightly more than a_{max} so that the body slips. At the instant the body begins to slip, the driver of the truck reduces the acceleration at a constant rate C. What must be the value of C such that the slipping stops just before the body is about to fall from the truck.

[Ans: 2.4 m/s^2, 0.2 m/s^3]

12. If the truck in the above problem is accelerating down an inclined plane 15° to the horizontal, (a) what is the minimum acceleration for which the body does not slip? (b) Is there a maximum limit to this acceleration?

[Ans: 0.27 m/s^2, 4.9 m/s^2]

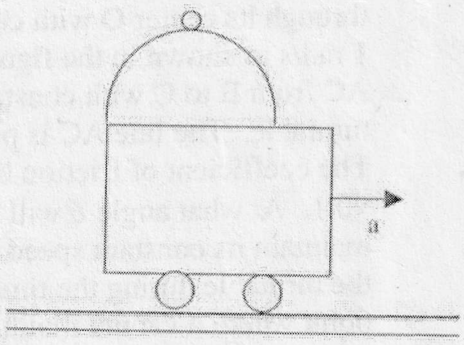
*13.A train is moving on a straight track with an acceleration of 2.5 m/s^2. A projectile is thrown from the floor of the train at an elevation of 45° and with an initial velocity of 4 m/s with respect to the train. The projectile is found to hit the floor at a distance of 1.6 m from the point on the train from which it was thrown. Determine the angle between the position vector of the projectile when it returns to the floor and the direction of motion of the train.

[Ans: 98°]

14. A train is moving with acceleration $\boldsymbol{a}$. A man on the train throws a ball with velocity v_0 relative to the train at an angle θ with respect to the vertical direction. The horizontal component of the velocity of the ball is in the same direction as $\boldsymbol{a}$. Calculate the angle θ, if the man catches the ball back.

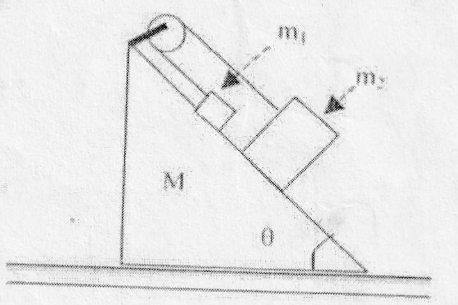
[Ans: $\tan\theta = a/g$]

15. A vertical frictionless semicircular track of radius 1 m is fixed on the edge of a movable trolley as shown in the figure. Initially the system is at rest and a mass m is kept at the top of the track. The trolley starts moving to the right with a uniform acceleration of $\frac{2}{9}g$. The mass slides down the track (without rotation), eventually losing contact with it. Calculate the angle at which the mass loses contact with the track.



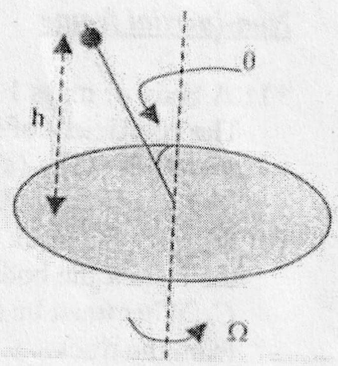
[Ans: 36.9°]

16. In the adjacent figure, the masses are $m_1 = 1$ kg, $m_2 = 3$ kg and $M = 9$ kg. The angle $\theta = 45°$. All the surfaces are frictionless. Find the acceleration of mass M.

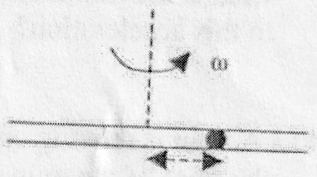


[Ans: 0.04 g]

*17.A long straight rigid wire of uniform cross-section is fixed at the center of a horizontal disc such that it makes an angle θ with the vertical as shown in the figure. A bead of mass m is held on the wire at height h from the plane of the disc. The coefficient of friction between the wire and the bead is μ. The disc rotates at a constant angular velocity Ω about the vertical axis passing through its center and the bead is released from its rest position. Find the maximum and the minimum values of Ω for which the bead is stationary on the wire at its initial position.



18. A particle of mass m is constrained to move along a frictionless groove, which is being rotated about a vertical axis through its center with angular velocity ω (see figure). If it starts at a distance R from the center at $t = 0$, what is the velocity at time t? Calculate the force, the groove walls exert on the mass in the horizontal direction.

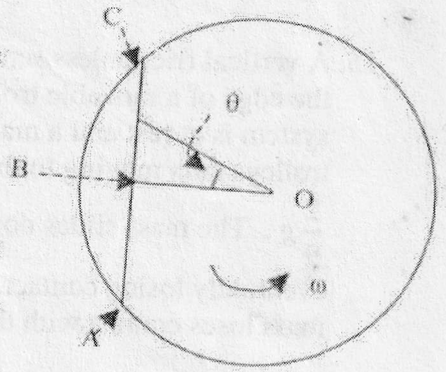


[Ans: $\dot{r} = \frac{R\omega}{2}(e^{\omega t} - e^{-\omega t})$; $\vec{N} = 2m\omega^2 \frac{R}{2}(e^{\omega t} - e^{-\omega t})\hat{\theta}$]

19. A small coin is places on a horizontal turntable at a distance of 10 cm from the center. The turntable, which is initially at rest, is given an angular acceleration of 10 rad/s^2. The coefficient of friction between the turntable and the coin is 0.5. Determine the angular distance traveled by the turntable before the coin begins to slip.

[Ans: 2.4 rad]

*20.A horizontal turntable is rotating about a vertical axis passing through its center O with constant angular velocity ω of magnitude 1 rad/s as shown in the figure. An insect is moving along the line AC from B to C with constant speed of 2 m/s with respect to the turntable. The line AC is perpendicular to OB, with OB = 1 m. The coefficient of friction between the insect and the turntable is √0.1. At what angle θ will it no longer be possible for the inscet to maintain its constant speed along AC. What is the angle turned by the turntable during the time in which inscet moves from B to the point where it cannot maintain its constant velocity.



[Ans $\theta = 45°$]